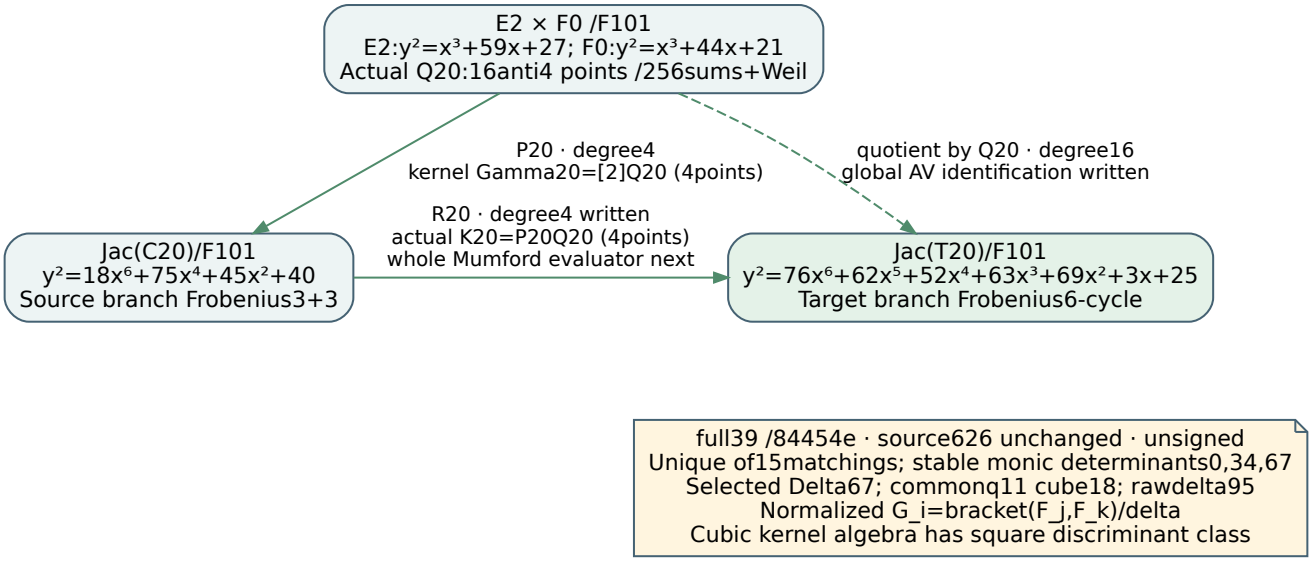
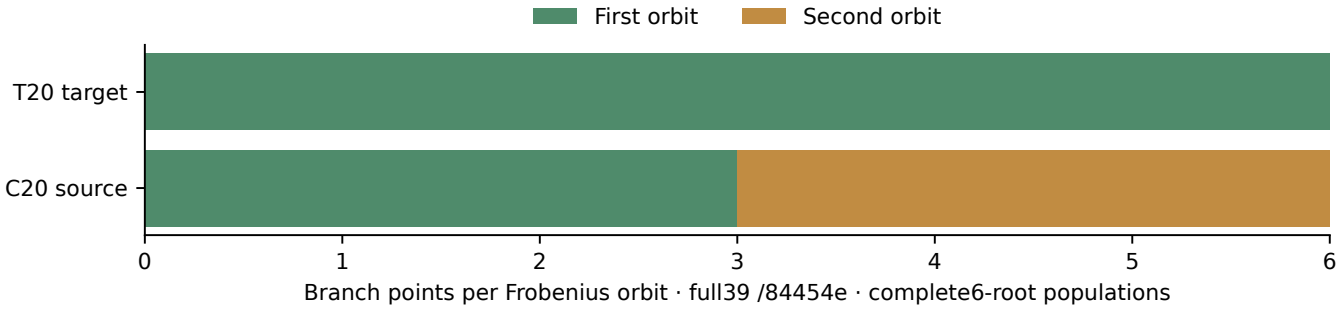
Actual T20 primitive-four kernel and raw target

2026-11-12 · geometry phaseA · full39 /84454e · unsigned

Complete39 native tests pass,zero failed/ignored/filtered;old reference+legacy guards,fmt,strictClippy and all626 input hashes pass. Geometry6141-byte JSON is byte-identical to diagnostic296872;old24475-byte kernel JSON matches9f7c80. Four bound local Lean passes give192written38modules/0trusted. Algebra01/d4220d passes23/10outputs/all13 hashes,old8 outputs byte-identical. Scientific standing remains unsigned. [pcc:unverified]



Actual graph/kernel/matching/target geometry;global quotient and degree use the written AV/Picard bridge. Whole new Mumford correspondence remains phaseB. No cost inference. [pcc:unverified]



Complete six-branch-point Frobenius orbits,not sampled or measured costs. Source3+3 becomes target6-cycle;all roots are retained. [pcc:unverified]

The actual P20Q20 kernel has4 valid two-torsion classes,all16 sums and Frobenius permutation[0,2,3,1]. It selects the unique matching [[0,4],[1,3],[2,5]] among15. Three stable matchings have monic determinants 0,34,67;selected67 is nonsingular. q=11,q³=18 gives rawdelta95 and the actual raw T20 equation above,smooth with N1=120,N2=10436,J12317. External CT49 shift2 is independently confirmed at this geometry scope. [pcc:unverified]

Prime-field quotient and normalization proof

Use common leading q in every F_i=q·Q_i,with productF_i equal to the raw C20 polynomial. The factors form a Galois-stable unordered splitting. For monic Delta67,the scaled determinant is delta=q³Delta=95. Normalized G_i=bracket(F_j,F_k)/delta yields productB_i/(18Delta³),the old native prototype convention. The generic genus2 codomain delta·product(brackets) differs by the square delta⁴;we retain the normalized coordinate throughout. No point count selects a retwist. Target leading76=50² is retained. [pcc:unverified]

The nonzero kernel points form a Frobenius three-cycle,so their coordinate algebra L is F101³. Its discriminant square class is1 because the sign of a three-cycle is positive. Bruin–Doerksen Proposition4.3 then identifies the F101 quotient with the actual normalized T20 model. Source smoothness, common cube leading,nonzero determinant and nonsplit twist hypotheses are checked at their exact scope. Thus a global separable degree4 (2,2) isogeny R20 has kernelK20 and adjoint norm[2]. Odd109/113 DLP subgroups are preserved;proper native whole-map execution remains phaseB. [pcc:unverified]

Faithful formal and checker scope

PRIMITIVE_FOUR_T20_GEOMETRY_PROOF.md/b24be8 gives actual kernels, quotient compatibility,scalar convention,BD discriminant twist and all written geometric bridges. Exact local Lean bindings89e906/02c312/fee2b1/ 151a58 cover determinant scale,normalization factorization,unit cancellation and actual Fin101 constants. Project192written38modules;gate4/0trusted. New IDC1h11c6b82bbe669b8f and IDC1h3147241afb3d4b6b use the unchanged checker32+64 points at fixed2^61-1,not F101. All10 RHS+1 controls reject. [pcc:unverified]

Whole134348-byte native output→standardXZ11468,with raw/archiveSHA256, source/start/finish/ exit/626afterhashes/both JSON comparisons and exact roundtrip. The6141-byte geometry JSON contains actual graph,field,kernel, Frobenius,branch,matching,scale,factor and target data. Compile017c05b4 is preserved in the separate failure report;the new-child-only fix changes no old type or visibility. [pcc:unverified]

LocalLinuxx86_64development checks,no benchmark isolation. Rust1.99.1/ Lean4.35.0. Reproduce full39/ oldguards and algebra23/10/38proofmodules. New218 compiled expected,not own-CI observed. Prior d87 CI succeeds 25/37/23/seven/209compiled183written36;published30fa CI43283121414 pending at38/188written37,check19:40UTC. Consumer1513 pins older338d35. [pcc:unverified]

Sources:BD0902.3480§4/nativeRic+genericengines/128eb1/afd164/063773/ external CT49f7d4c6be. KB unavailable,actual fallback receipts persistent, no novelty inference. Catalog72 count unchanged. Next:proper polynomial fibres/ramified/repeated/infinity/basepoint/dual2/Frobenius/all12317/DLP/ actual C0→C20→T20 chain,plus extra16vs4 geometric parity controlb0c5f6. No phaseB result or measured cost advantage claimed. REPORT.md links full source/custody/editable graph/data/renderer and inspected PDF. [pcc:unverified]